CSE Graduate Seminar

Log-Structured Filesystem in the Era of AI

Professor Youjip Won
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TIME & LOCATION

📅 Date: Mar 4 (Wed), 2026
🕒 Time: 16:00 – 17:00
📍 Location: Chemistry Bldg, Room 330118

ABSTRACT

In this talk, we will review how the log-structured filesystem evolves in the era of the AI. Log-structured filesystem is write optimized filesystem that have primarily been proposed to efficiently handle the random write workload from the application. It clusters the metadata and data together and transforms the random writes into a sequential one. Despite its benefits, it had not well adopted in the production environment. The prime reason is that the log-structured filesystem needs to consolidate the valid blocks and reclaims the invalidated filesystem regions. This activity is called garbage collection. During garbage collection, the filesystem blocks all foreground write requests. For the past years, we developed a number of techniques that mitigate or eliminate the garbage collection overhead from the host. In this talk, we will mainly deal with two issues. First, we will explain a number of ways to mitigate the filesystem garbage collection overhead from the host. Second, we will discuss the performance limitations which the modern log-structured filesystem faces.

BIO

Prof. Won is ICT Endowed Chair Professor at School of Electrical Engineering, KAIST. He was a dept. head and a director of Center for File and Storage Systems Technology (CFSR) at Hanyang University. He moved to KAIST after 20 years of tenure at Hanyang University. His work on IO stack optimization in Android platform identified the root cause of the IO inefficiencies in the Android smartphones (USENIX ATC 2013, best paper award). His work on IO stack design on the Flash Storage proposes a new way of designing the filesystem, block layer and device firmware for the Flash storage (USENIX FAST 2018, best paper award). He is known for his work on the Android IO stack optimization, filesystem and block layer design for SSD and NVRAM. His research interests include Operating System, Distributed System, Storage System and Software support for byte-addressable NVRAM.